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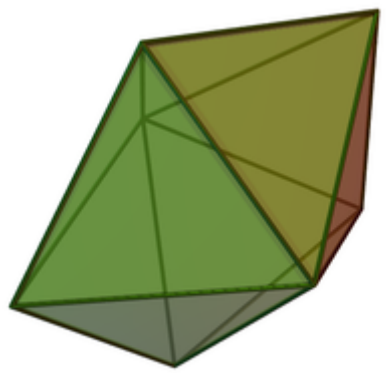
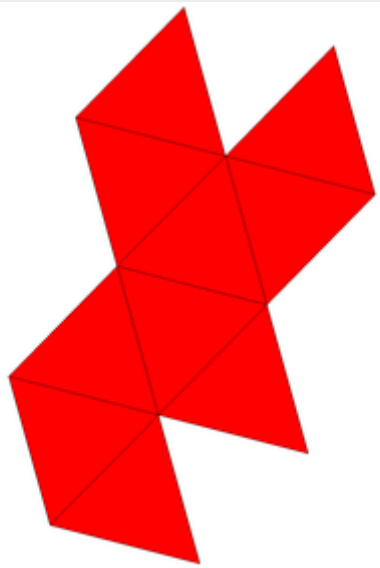
Pentagonal bipyramid

In geometry, the **pentagonal bipyramid** (or **pentagonal dipyramid**) is a polyhedron with 10 triangular faces. It is constructed by attaching two pentagonal pyramids to each of their bases. If the triangular faces are equilateral, the pentagonal bipyramid is an example of deltahedra, and of Johnson solid.

The pentagonal bipyramid may be represented as 4-connected well-covered graph. This polyhedron may be used in the chemical compound as the description of an atom cluster known as pentagonal bipyramidal molecular geometry.

Pentagonal bipyramid	
	
Type	Bipyramid, Johnson $J_{12} - \boldsymbol{J_{13}} - J_{14}$
Faces	10 triangles
Edges	15
Vertices	7
Vertex configuration	$5 \times (3^4) + 2 \times (3^5)$
Symmetry group	D_{5h}
Dual polyhedron	pentagonal prism
Properties	convex, face-transitive
Net	
	

Construction

Like other bipyramids, the pentagonal bipyramid can be constructed by attaching the base of two pentagonal pyramids.[1] These pyramids cover their pentagonal base, such that the resulting polyhedron has 10 triangles as its faces, 15 edges, and 7 vertices.[2] The pentagonal bipyramid is said to be right if the pyramids are symmetrically regular and both of their apices are on the line passing through the base's center; otherwise, it is oblique. If the pyramids are regular, then all edges of the triangular bipyramid are equal in length, making up the faces equilateral triangles. A polyhedron with only equilateral triangles as faces is called a deltahedron.[3] There are only eight different convex deltahedra, one of which is the pentagonal bipyramid with regular faces. More generally, the convex polyhedron in which all faces are regular is the Johnson solid, and every convex deltahedra is a Johnson solid. The pentagonal bipyramid with the regular faces is among numbered the Johnson solids as $\boldsymbol{J_{13}}$, the thirteenth Johnson solid.[4]

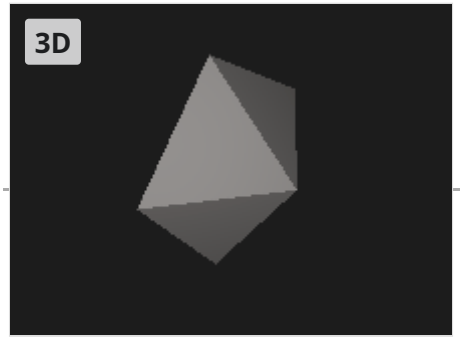
Properties

A pentagonal bipyramid's surface area is 10 times that of all triangles. In the case of edge length $\boldsymbol{a}$, its surface area is:[2]

$$\frac{5\sqrt{3}}{2}a^2 \approx 4.3301a^2.$$

Its volume can be calculated by slicing it into two pentagonal pyramids and adding their volume. In the case of edge length a, this is:[2]

$$\frac{1}{12}\left(5+\sqrt{5}\right)a^3 \approx 0.603a^3.$$



Johnson solid J_{13}

The pentagonal bipyramid has three-dimensional symmetry group of dihedral group D_{5h} of order 20: the appearance is symmetrical by rotating around the axis of symmetry that passing through apices and base's center vertically, and it has mirror symmetry relative to any bisector of the base; it is also symmetrical by reflecting it across a horizontal plane. The dihedral angle of a pentagonal bipyramid with regular faces can be calculated by adding the angle of pentagonal pyramids. The dihedral angle of a pentagonal pyramid between two adjacent triangles is approximately $138.2°$, and that between the triangular face and the base is $37.4°$. Therefore, the dihedral angle of a pentagonal pyramid with regular faces between two adjacent triangular faces, on the edge where two pyramids are attached, is $74.8°$.[5]

The pentagonal dipyramid is 4-connected, meaning that it takes the removal of four vertices to disconnect the remaining vertices. It is one of only four 4-connected simplicial well-covered polyhedra, meaning that all of the maximal independent sets of its vertices have the same size. The other three polyhedra with this property are the regular octahedron, the snub disphenoid, and an irregular polyhedron with 12 vertices and 20 triangular faces.[6]

The dual polyhedron of a pentagonal bipyramid is the pentagonal prism. This prism has 7 faces: 2 pentagonal faces are the base, and the rest are 5 rectangular faces. More generally, the dual polyhedron of every bipyramid is the prism.

Applications

In the geometry of chemical compounds, the pentagonal bipyramid can be used as the atom cluster surrounding an atom. The pentagonal bipyramidal molecular geometry describes clusters for which this polyhedron is a pentagonal bipyramid. An example of such a cluster is iodine heptafluoride in the gas phase.[7]

Pentagonal bipyramids and related five-fold shapes are found in decahedral nanoparticles,[8] which can also be macroscopic in size when they are also called fiveling cyclic twins in mineralogy.[9]

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